MHF® Series - Micro RF Coaxial Connectors

Click the photos below to get more detailed product information.

Product name		MHF® I LK	MHF® I	MHF® I	MHF® III
Appearance					
		MHF I with mechanical lock	**Best Insertion Loss**	**Most Cable O.D. options**	**low profile 1.60 mm**
Plug part number		-	20767-001R	20278-112R-**	20609-002R
Receptacle part number		20279-001E-** (3 pads) 20441-001E-01 (4 pads)			20369-001E-**
Maximum height (mm)		2.9	3.0	2.5	1.6
Outside dimension of receptacle (mm)		3.0 x 3.0			2.0 x 2.0
Coax O.D. (Center Conductor AWG)	2.00 mm (26)		●		
	1.80 mm (30)	●		●	
	1.37 mm (30)	●		●	
	1.32 mm (32)	●		●	
	1.13 mm (32)	●		●	
	0.95 mm (33)				
	0.81 mm (33)				
	0.81 mm (36)	●		●	●
	0.64 mm (36)				●
	0.48 mm (38)				
Frequency		DC - 9 GHz			DC - 6 GHz
VSWR (L=100mm)	DC - 3GHz	1.3 max.(PLUG) /1.3 max.(RECE)			1.3 max.
	3GHz - 6GHz	1.5 max.(PLUG) /1.4 max.(RECE)			1.5 max.
	6GHz - 9GHz	1.9 max.(PLUG) /1.8 max.(RECE)		1.6 max.(PLUG) 1.8 max.(RECE)	-
	9GHz - 12GHz	-			
	12GHz - 15GHz	-			
Service temp. (Celsius)		-40 degree - 90 degree			
Characteristic impedance		50ohm			
Rated voltage		AC60V			
Contact resistance		20m ohm max.			
Withstand voltage		AC200V/min			
Insulation resistance		500M ohm min./DC100V			

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MHF® Series - Micro RF Coaxial Connectors

Click the photos below to get more detailed product information.

Product name		MHF® 4L LK	MHF® 4L	MHF® 4L	MHF® 4L	MHF® 4
Appearance						
		MHF 4L with Mechanical lock	**Best Insertion Loss and VSWR**	**Industry Standard for M.2**	**Industry Standard for M.2**	**low profile1.20 mm**
Plug part number		-	20632-001R-37	20565-001R-13	20572-001R-08	20611-001R
Receptacle part number		20449-001E-**(MHF 4) 20579-001E(MHF 4L)				
Maximum height (mm)		2.0	1.7	1.4	1.2	1.2
Outside dimension of receptacle (mm)		2.0 x 2.0				
Coax O.D. (Center Conductor AWG)	2.00 mm (26)					
	1.80 mm (30)					
	1.37 mm (30)	●	●			
	1.32 mm (32)					
	1.13 mm (32)			●		
	0.95 mm (33)			●		
	0.81 mm (33)					
	0.81 mm (36)				●	●
	0.64 mm (36)				●	
	0.48 mm (38)					
Frequency		DC – 12 GHz (MHF 4L Plug x MHF 4L Rece) , DC – 9 GHz (MHF 4L Plug x MHF 4 Rece)				DC - 9 GHz
VSWR (L=100mm)	DC - 3 GHz	1.3 max.				
	3 GHz - 6 GHz	1.4 max.		1.45 max		1.5 max.
	6 GHz - 9 GHz	1.5 max.		1.6 max.		-
	9 GHz - 12 GHz	1.6 max.		1.9 max.		-
	12 GHz - 15 GHz	-				
Service temp. (Celsius)		-40 degree - 90 degree				
Characteristic impedance		50ohm				
Rated voltage		AC60V				
Contact resistance		20m ohm max.				
Withstand voltage		AC200V/min				
Insulation resistance		500M ohm min./DC100V				

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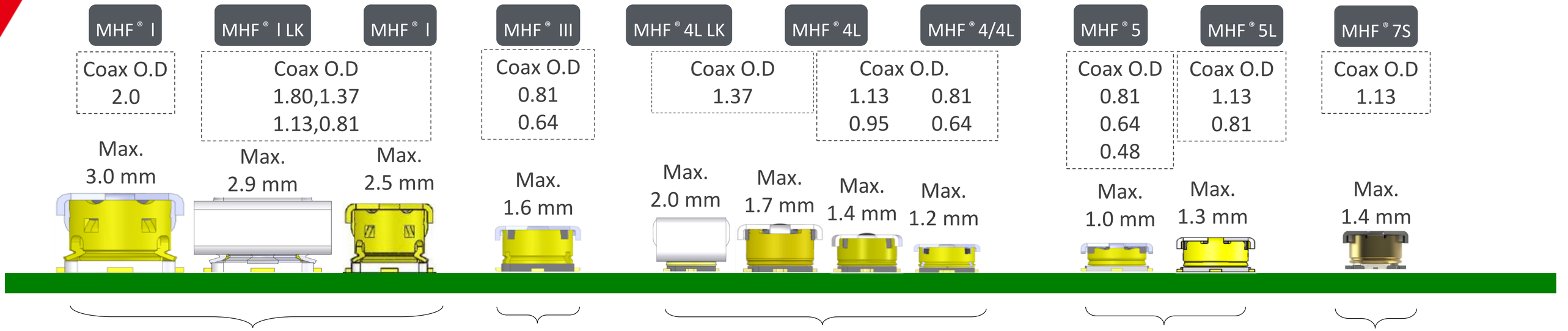
MHF® Series - Micro RF Coaxial Connectors

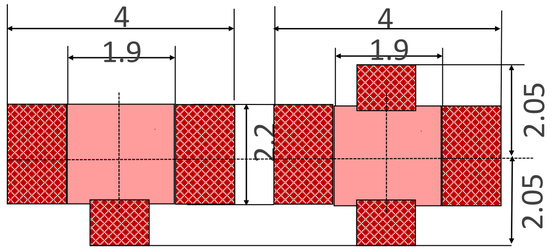
Click the photos below to get more detailed product information.

Product name		MHF® 5L	MHF® 5L	MHF® 5	MHF® 5	MHF® 7S
Appearance						 ZenShield®
		Improved Insertion Loss and VSWR through 15GHz	**Improved Insertion Loss and VSWR through 15GHz**	**low profile 1.00 mm**	**low profile 1.00 mm**	**Unique shielded design increases EMC performance**
Plug part number		20668-001R-13	20714-001R-81	20711-001R-81	20615-002R-48	20980-001R-13
Receptacle part number		20566-001E-01				20981-001E-02
Maximum height (mm)		1.3		1.0		1.4
Outside dimension of receptacle (mm)		2.0 x 2.0				2.0 x 2.0
Coax O.D.(Center Conductor AWG)	2.00 mm (26)					
	1.80 mm (30)					
	1.37 mm (30)					
	1.32 mm (32)					
	1.13 mm (30)					
	1.13 mm (32)	●				●
	0.95 mm (33)					
	0.81 mm (33)		●			
	0.81 mm (36)			●		
	0.64 mm (36)			●		
	0.48 mm (38)				●	
Frequency		DC - 15GHz		DC - 12GHz		DC - 15GHz
VSWR (L=100mm)	DC - 3GHz	1.3 max.				1.3 max.
	3GHz - 6GHz	1.4 max.		1.5 max.		1.35 max.
	6GHz - 9GHz	1.5 max.		1.6 max.		1.4 max.
	9GHz – 12GHz	1.5 max.		1.7 max.	1.6 max.	1.45 max.
	12GHz - 15GHz	1.6 max.		-		1.5 max.
Service temp. (Celsius)		-40 degree - 90 degree				
Characteristic impedance		50ohm				
Rated voltage		AC60V				
Contact resistance		20m ohm max.				
Withstand voltage		AC200V/min				
Insulation resistance		500M ohm min./DC100V				

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Micro RF coax connector : Foot pattern



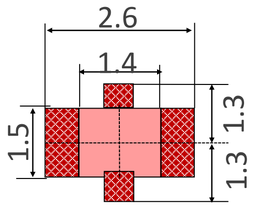


MHF® I

Rece 3Pad

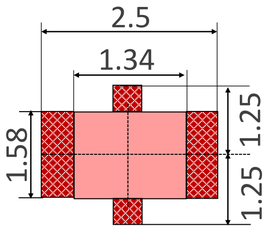
MHF® I

Rece 4Pad



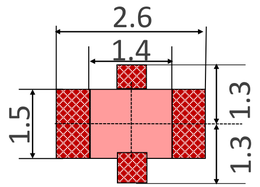
MHF® III

Rece



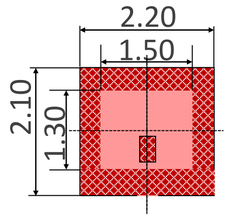
MHF® 4/4L

Rece



MHF® 5/5L

Rece



MHF® 7S

Rece

Confidential C

Accessory overview

Product					MHF® I	MHF® I LK	MHF® III	MHF® 4	MHF® 4L	MHF® 4L LK	MHF® 5	MHF® 7S
SMA adapter		Durability (cycles)										
	for Plug	for electrical performance check	3K		90193-001		90285-001	90449-001			90543-0001	91255-0001
				SMA Type	Plug		Plug	Jack			Plug	Jack
		for voltage check	10K		90194-001		90286-001	90449-003-01			90586-0001	91245-0001
	for Receptacle	for electrical performance check	500		90193-002		90285-002	90449-002			90543-0002	91256-0001
				SMA Type	Jack		Jack	Jack			Jack	Jack
		High cycle Normal	10K		90539-001		90491-001	90698-0001			90793-0001	-
		High cycle Narrow pitch Cable	100	 Cable Probe	90577-0100 (100mm) 90577-0350 (350mm)		-	90577-0100 (100mm) 90577-0350 (350mm)				91197-300*
		High cycle Narrow pitch Probe	10K		90575-0001		-	90576-0001			90847-0001	91145-0001
		High cycle Detachable Base	200K	 Base Probe	90684-0001							-
		High cycle Detachable Probe	10K		90416-001		-	90683-0001			90690-0001	-
Hand tool	Mating and unmating tool				90224-001 90885-0001 (for 20767-001R)	-	-	90435-001	90609-0001 (for 20572-001R-08 / 20565-001R-13) 90873-0001 (for 20632-001R-37)	-	90612-0001	91186-0001
	Unmating tool				90192-001	-	90287-001	-	-	-	-	-
	Mating tool				-	-	-	-	-	-	90624-0001	-
	Mating and unmating tweezer				-	-	90501-001	90482-001	90616-0008 90615-0013	-	-	-
	Locking and unlocking tool				-	91026-0001	-	-	-	91166-0001	-	-

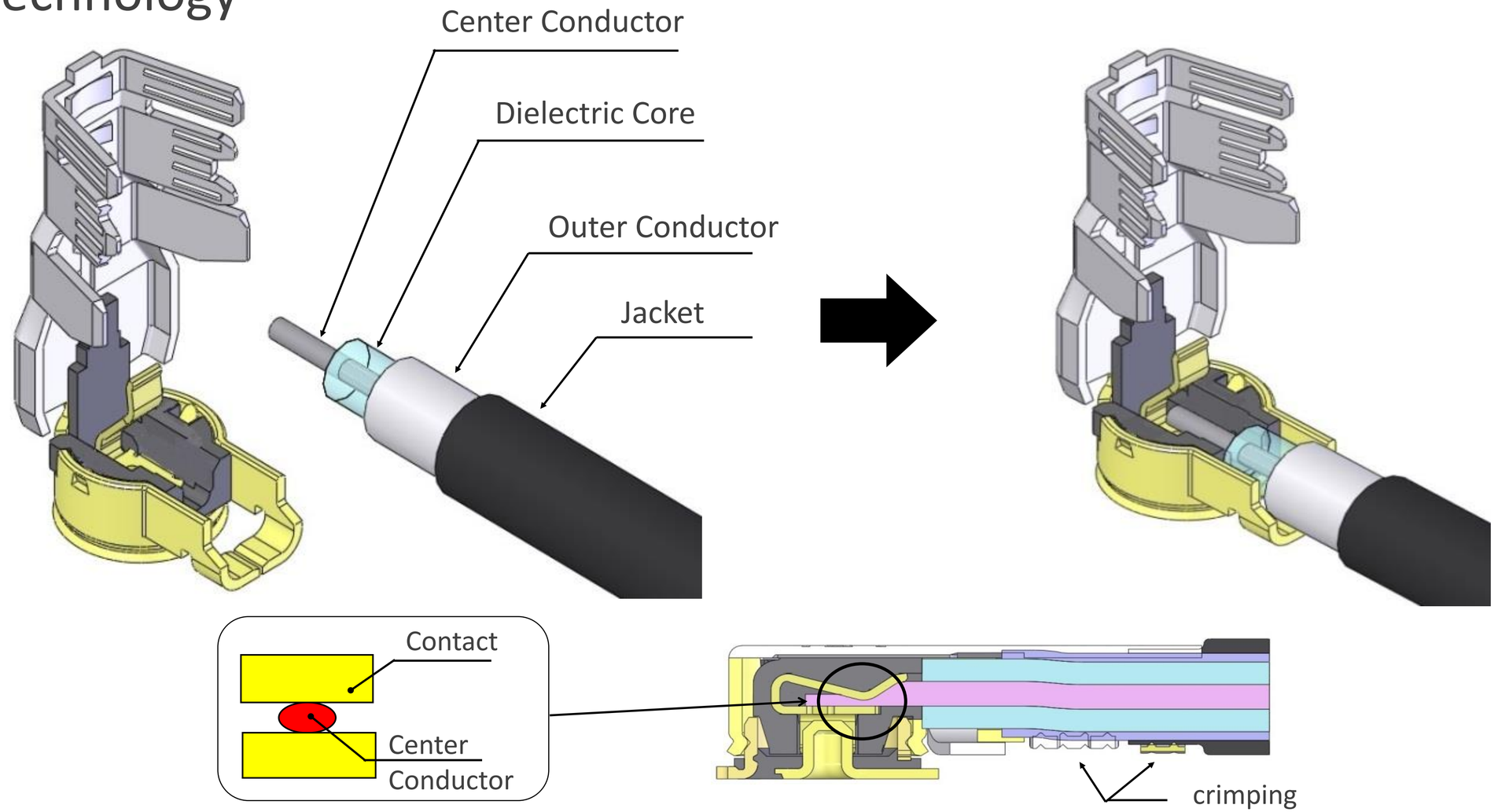
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MHF® Harnesses

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Micro RF coax connector : Cable crimp termination technology

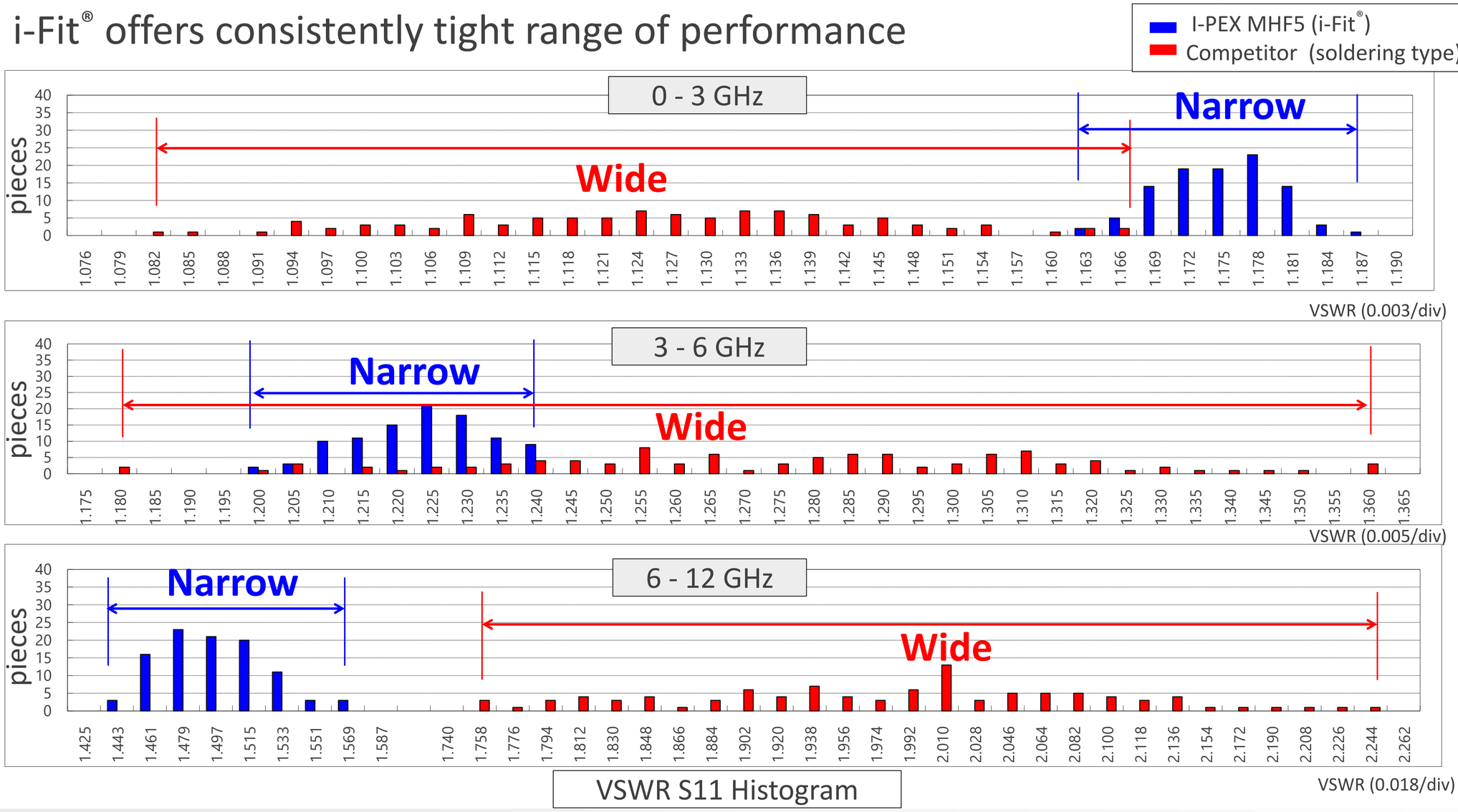
i-Fit® Technology



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Solder-less crimp “i-Fit®” advantage versus soldering types

i-Fit® offers consistently tight range of performance



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Solder-less crimp “i-Fit®” advantage versus manual soldering types

Significantly reduced tact time with i-Fit®

		I-PEX MHF "i-Fit®"	Manual Soldering
Tact time	Cable strip	Same tact time	
	Termination	**6** seconds/ea.	**30** seconds/ea.
Qty of cable assemblies a worker can make per month		105,600 assemblies	21,120 assemblies
# of operator needed	1M cable assemblies/mo	10	48
	2M cable assemblies/mo	19	95

*Calculated based on operation hours of 22days/mo (8hours/day)



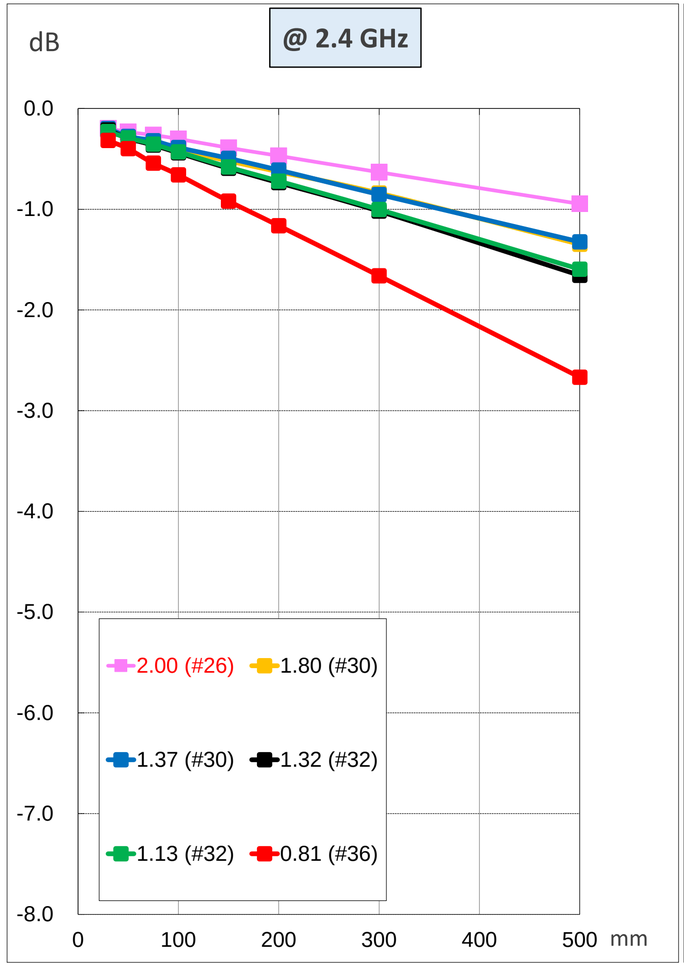
i-Fit® Semi-Auto crimp tool :

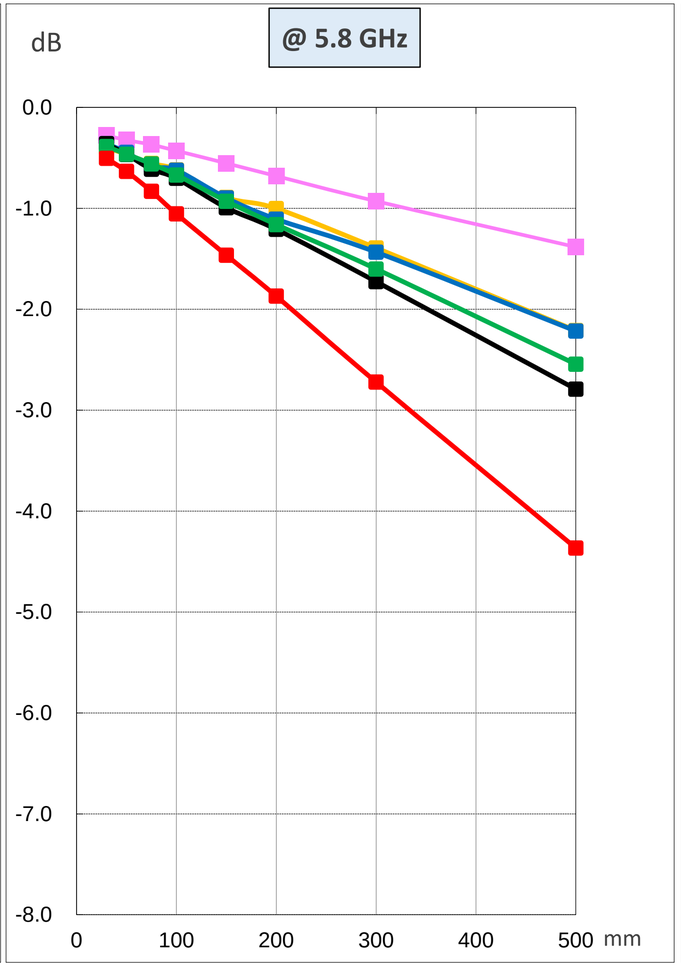
I-PEX has sold over 1,000 units of i-Fit crimping tool.
These are working in many antenna / harness makers around the world.

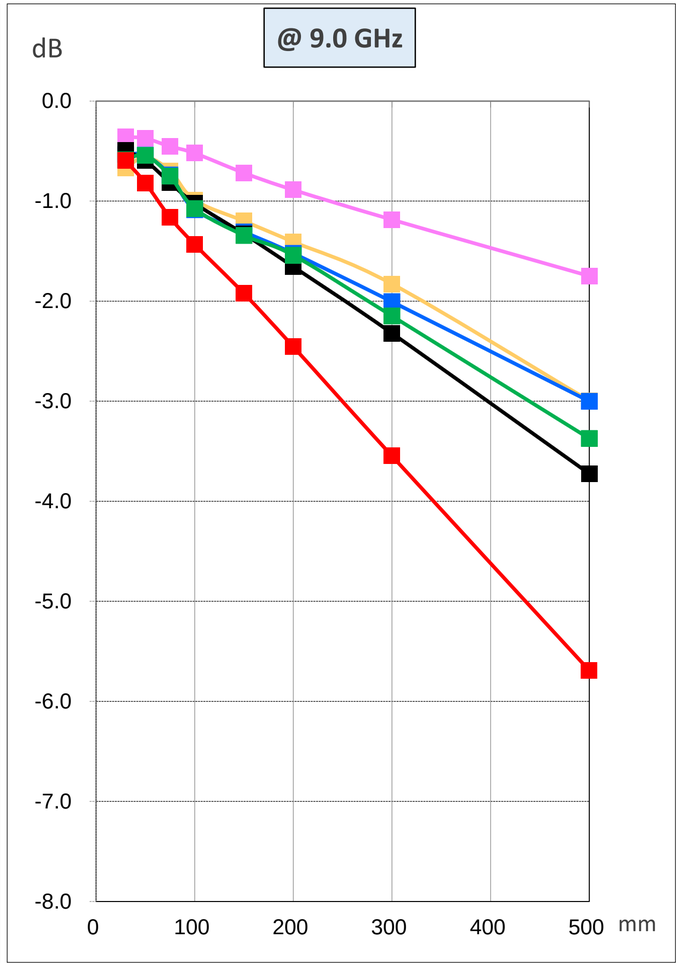
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Cable insertion loss (MHF® I)

Harness Length (mm) vs Insertion Loss (S21)



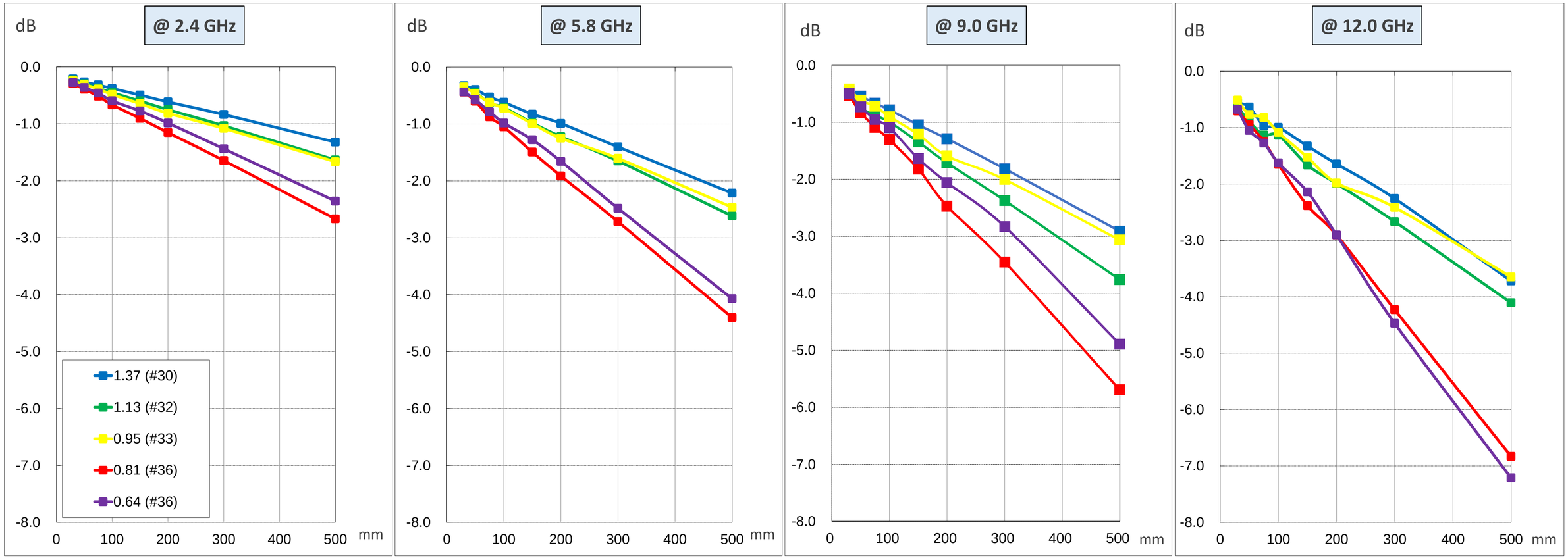




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Cable insertion loss (MHF® 4L)

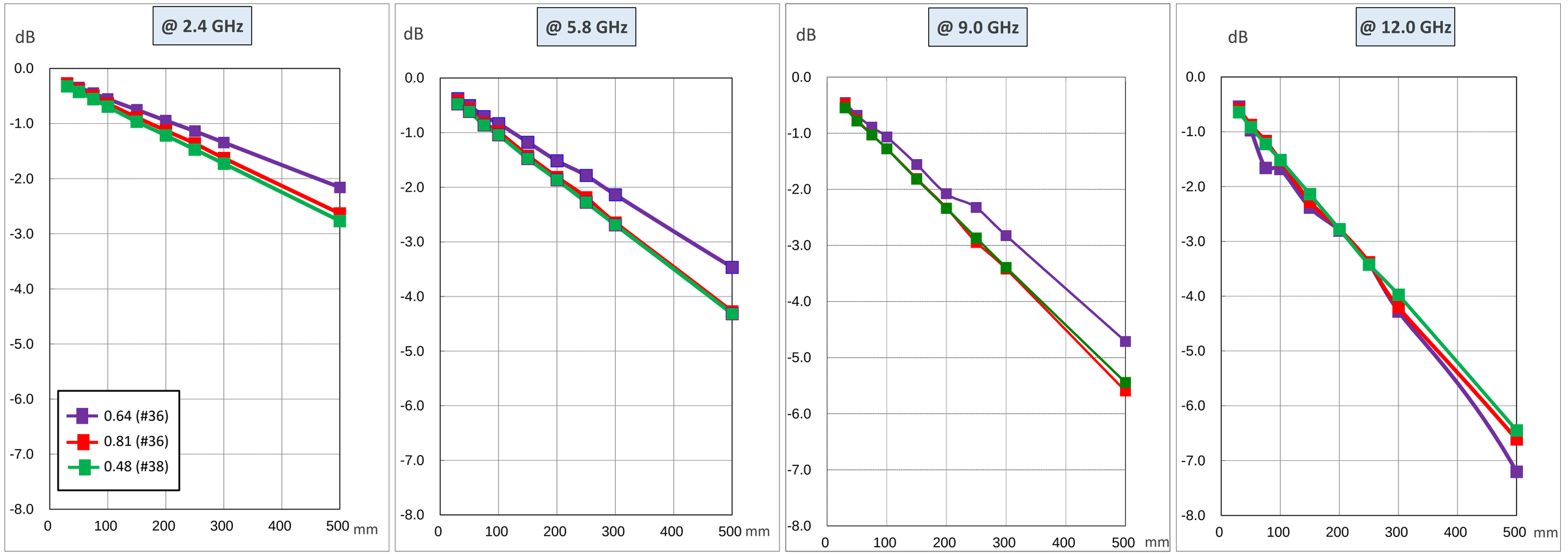
Harness Length (mm) vs Insertion Loss (S21)



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Cable insertion loss (MHF® 5)

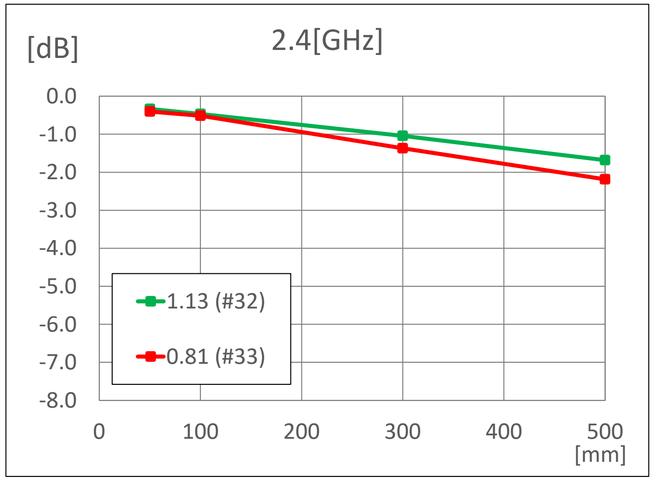
Harness Length (mm) vs Insertion Loss (S21)

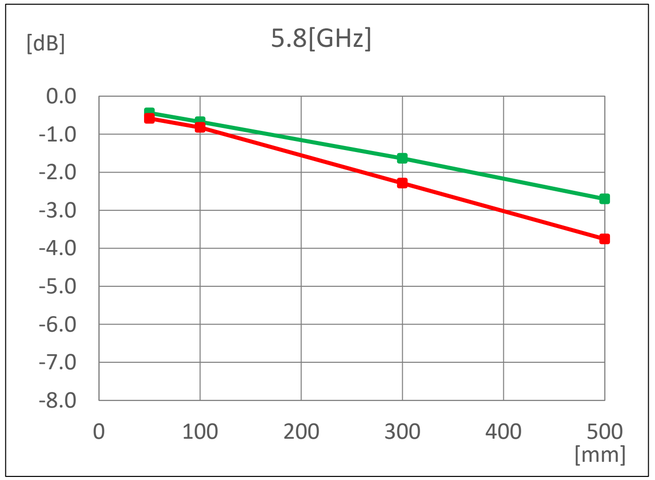


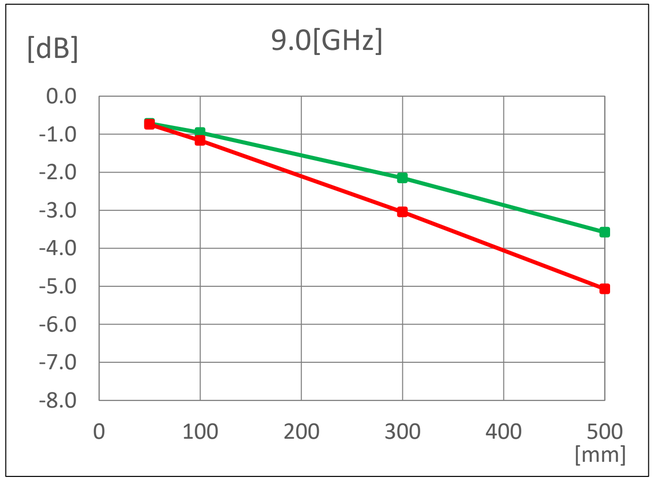
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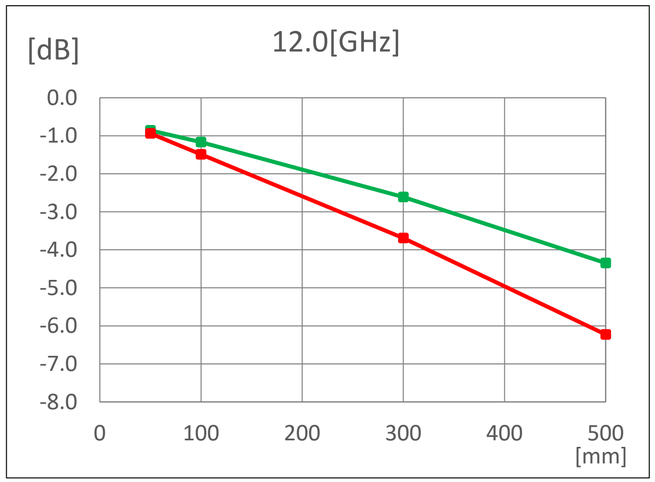
Cable insertion loss (MHF® 5L)

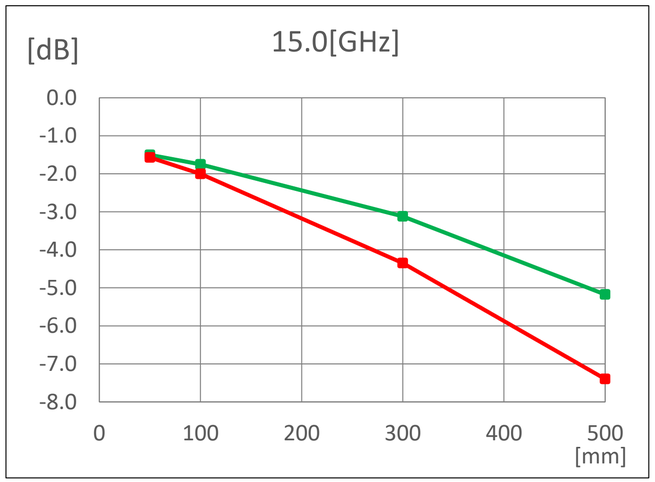
Harness Length (mm) vs Insertion Loss (S21)







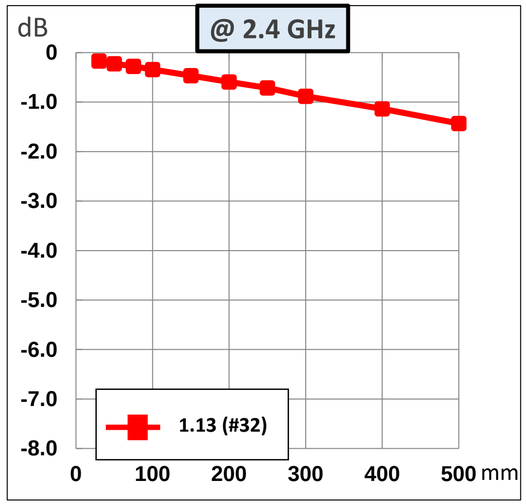


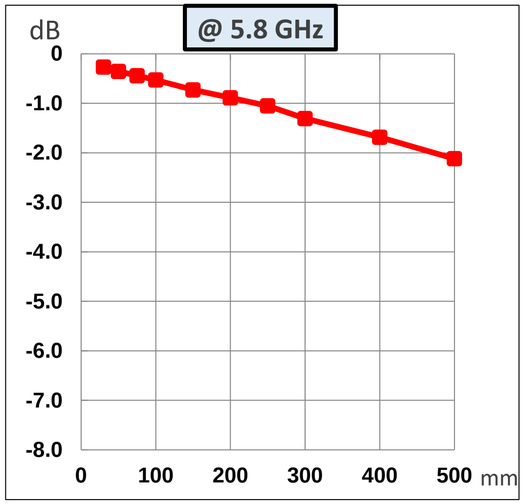


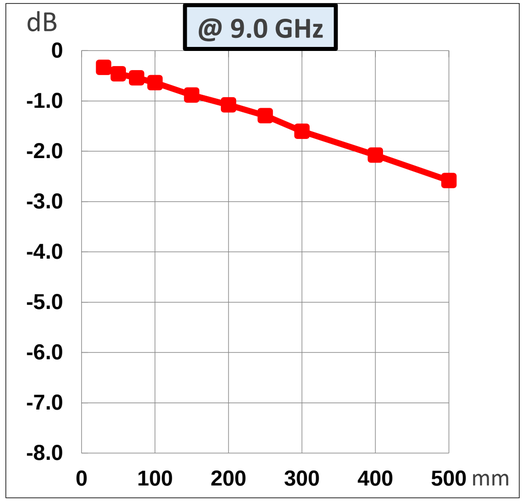
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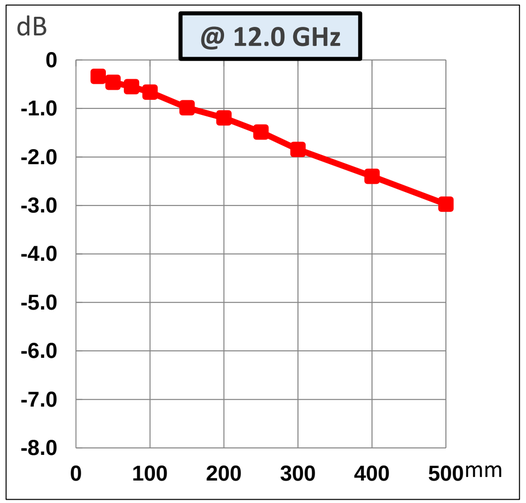
Cable insertion loss (MHF® 7S)

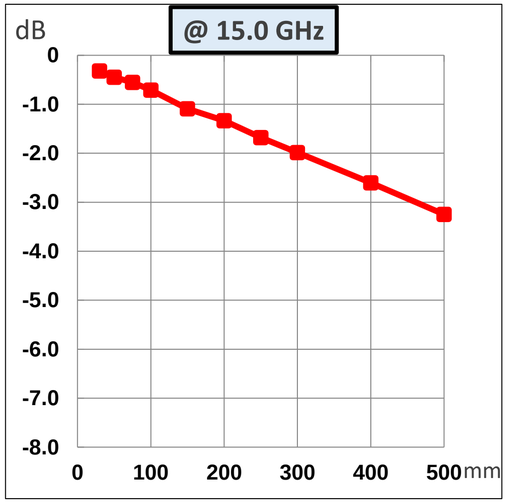
Harness Length (mm) vs Insertion Loss (S21)







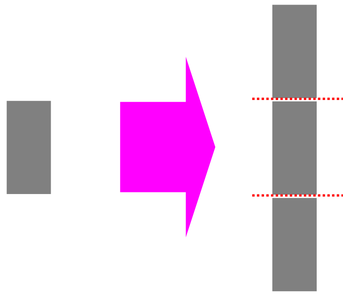




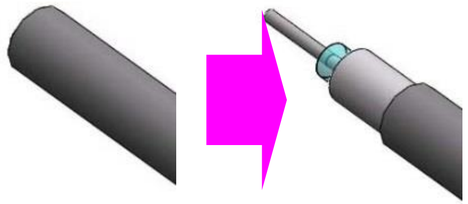
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Yes, I-PEX provides harness assemblies!
…And they are made with Fully Automated Machines and our patented i-Fit® technology!!!

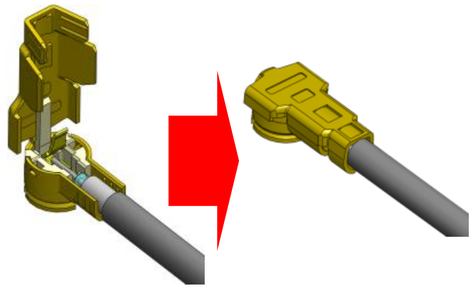
1 Cable cut



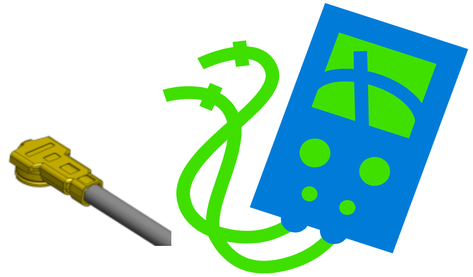
2 Cable strip



3 Crimp



4 Electric Inspection



Fully Automated Process



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Cable clamp and surface mount clip for grounding

		Grounding clamp 3mm length	Grounding clamp 6mm length
Appearance			
Coax O.D. (Center Conductor AWG)	1.37 mm (30)	●	●
	1.13 mm (32)	●	●
	0.81 mm (36)	●	●
	0.64 mm (36)	●	●

Clip for grounding clamp MP-A series

● MP-A04 (P/N 3224-0001)
● MP-A02 (P/N 3182-0001)
● MP-A01 (P/N 3096-0001)
● MP-A03 (P/N 3186-0001)

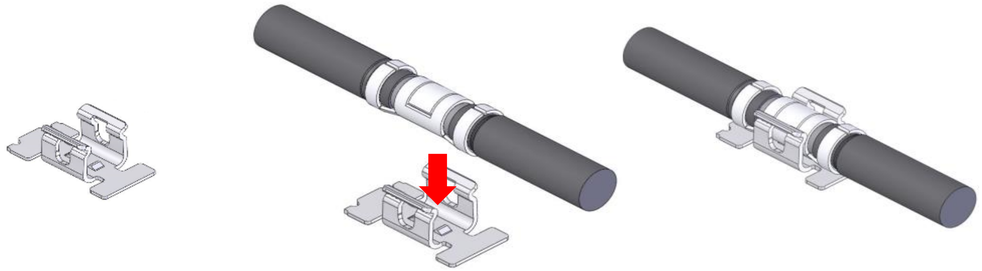
Minimum distance between clamps



A	B	C
12mm min.	10mm min.	10mm min.

*Depending on the harness specifications, this condition may not apply. Please contact our sales for details.

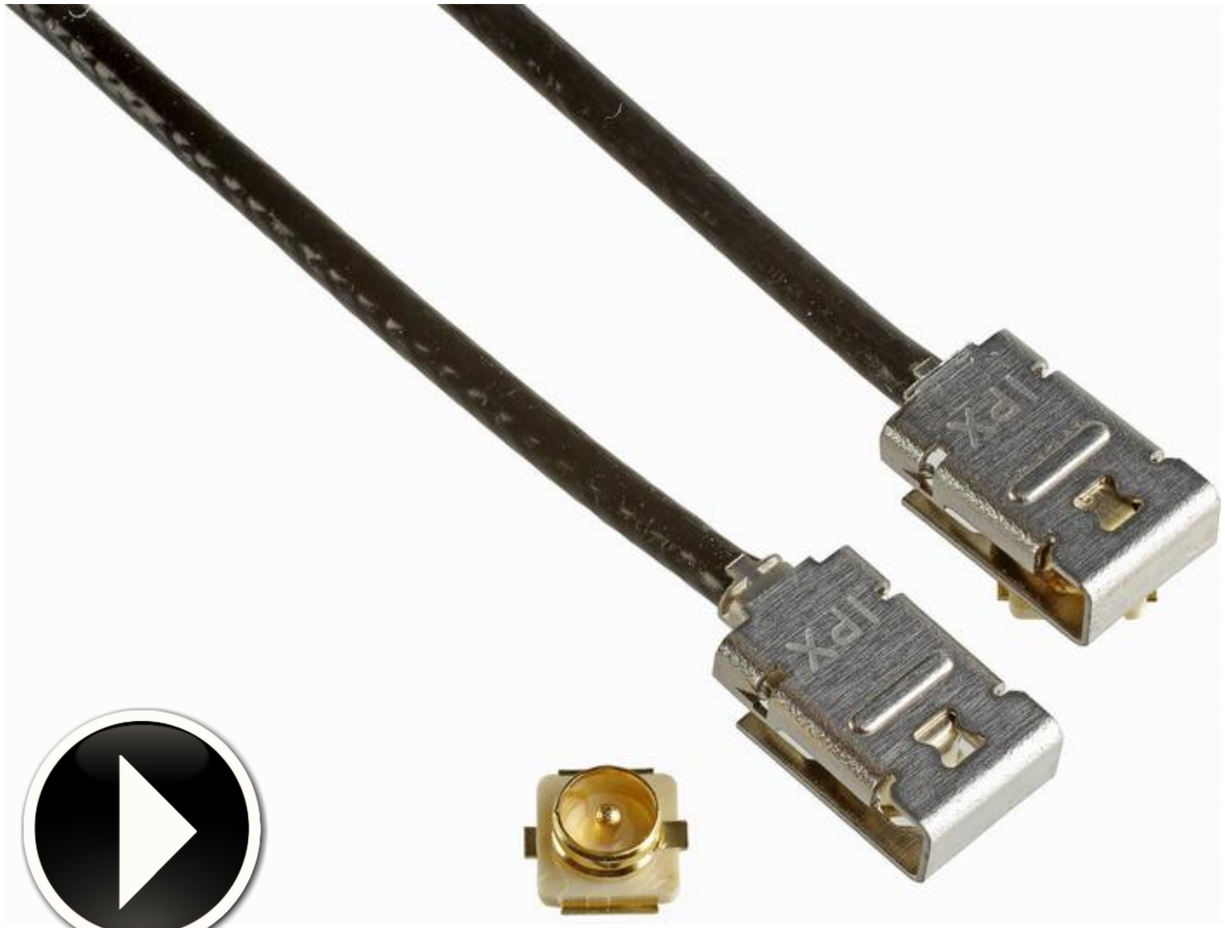
Clips are packaged in tape and reel for surface mount.



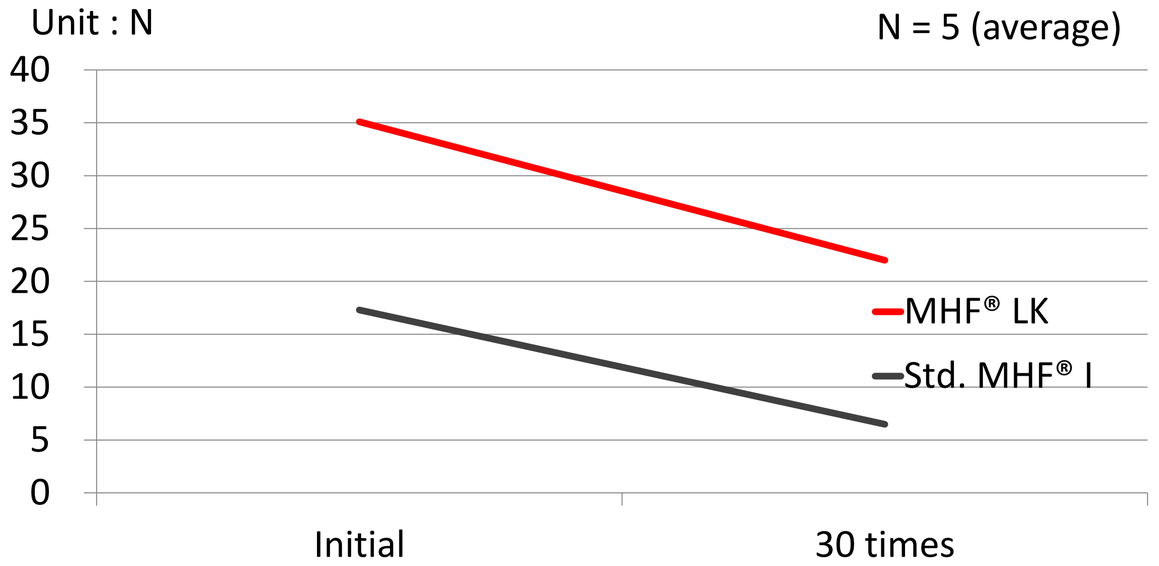
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MHF® I LK

Built-in locking function for MHF I



Disengagement Force Comparison



Series	MHF® I LK		Standard MHF® I	
	Initial	30 times*	Initial	30 times
SPEC.	20 MIN.	10 MIN.	5 MIN.	3 MIN.

* Reference Value

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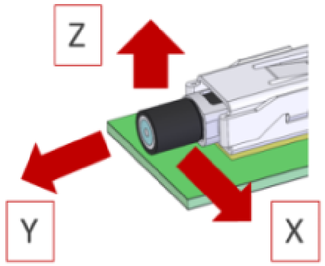
MHF®-TI

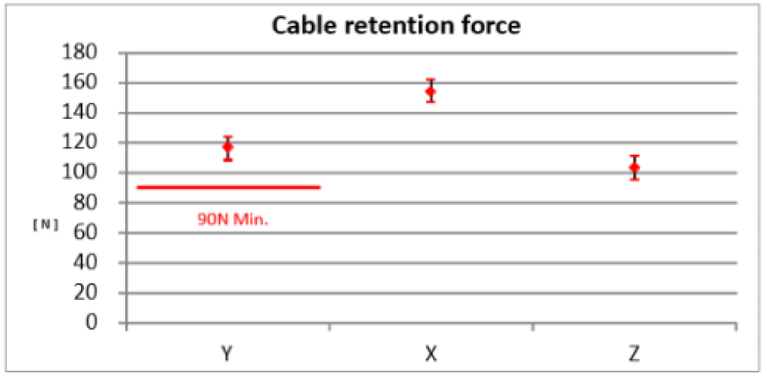
Click the photo below to get more detailed product information.

Product name		MHF® -TI
Appearance		
		Robust SMT connector for automotive telematics antenna
Plug part number		20859-001R-01
Receptacle part number		20860-001E-01
Maximum height (mm)		3.65
Outside dimension of receptacle (mm)		9.4 x 5.18
Coax O.D. (Center Conductor AWG)		3.00 mm (24)
		2.80 mm (26)
Frequency		DC - 6GHz
VSWR (L=100mm)	DC – 6GHz	1.5 max.
Service temp. (Celsius)		-40 degree ~ +105 degree
Characteristic impedance		50ohm
Rated voltage		AC60V
Contact resistance		20 m ohm max.
Withstand voltage		AC200V/min
Insulation resistance		500M ohm min./DC100V
Automotive Standards		USCAR-2 (*)

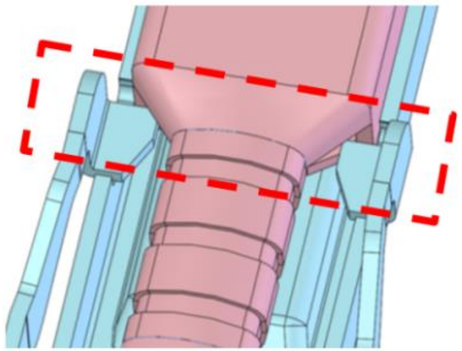
Note : (*) Rev.6 - 5.4.6 (Vibration), 5.6.1 (Thermal Shock), 5.6.2 (Temp. & Humidity Cycle), High Temp. Exposure & Life (5.6.3) evaluation passed

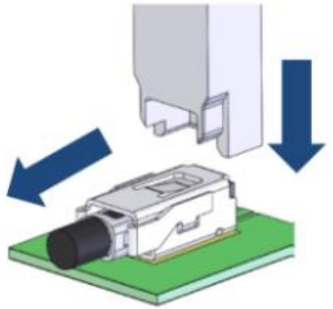
■ High retention force (90N minimum)

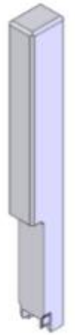




■ Permanent mechanical lock with tactile indication of secure contact







Un-Mating tool
91097-0001

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RF Switch

MHF®-SW Series

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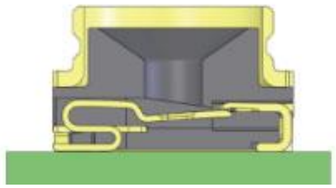
RF Switch: MHF®-SW23 & Accessories

Click the photos below to get more detailed product information.

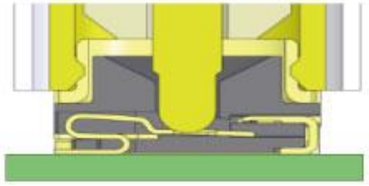
Product name		MHF®-SW23
Feature		
		Contaminant discontinuity resistant design **2.3 x 2.3 mm footprint**
Part number		20549-001E/-01
Dimension (mm)		2.3 x 2.3 x 1.35 (H)
Rated power		2W max.
Frequency		DC - 11.0GHz
V.S.W.R.	DC - 3.0GHz	1.2 max. (DC - 2.5GHz)
	3.0GHz - 6.0GHz	1.3 max. (2.5 - 6.0GHz)
	6.0GHz - 11.0GHz	1.5 max.
Insertion loss	DC - 3.0GHz	0.15 dB max. (DC - 2.5GHz)
	3.0GHz - 6.0GHz	0.20 dB max. (2.5 - 6.0GHz)
	6.0GHz - 11.0GHz	0.40 dB max.
Isolation	DC - 3.0GHz	20 dB Max. (DC - 2.5GHz)
	3.0GHz - 6.0GHz	15 dB max.
	6.0GHz - 11.0GHz	12 dB max.
Service temp. (Celsius)		-40 degree ~ +85 degree
Characteristic impedence		50 ohm
Contact resistance	Signal contact	100m ohm
	Ground contact	100m ohm
Insulation resistance		DC100 1000M ohm min :Initial 10M ohm min :After test
Withstand voltage		AC 100V: 1min

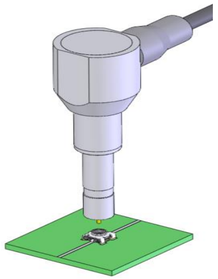
Product name				MHF®-SW23
Appearance				
Probe	Manual probe		Friction lock	90582-**** *(0300/0400)
			Mechanical lock	-
	Durability		-	1K Cycles
	High cycle: without floating			90604-0001
	Durability			10K Cycles
	High cycle: with floating			90605-0001
	Durability			200K Cycles

On State



Off State





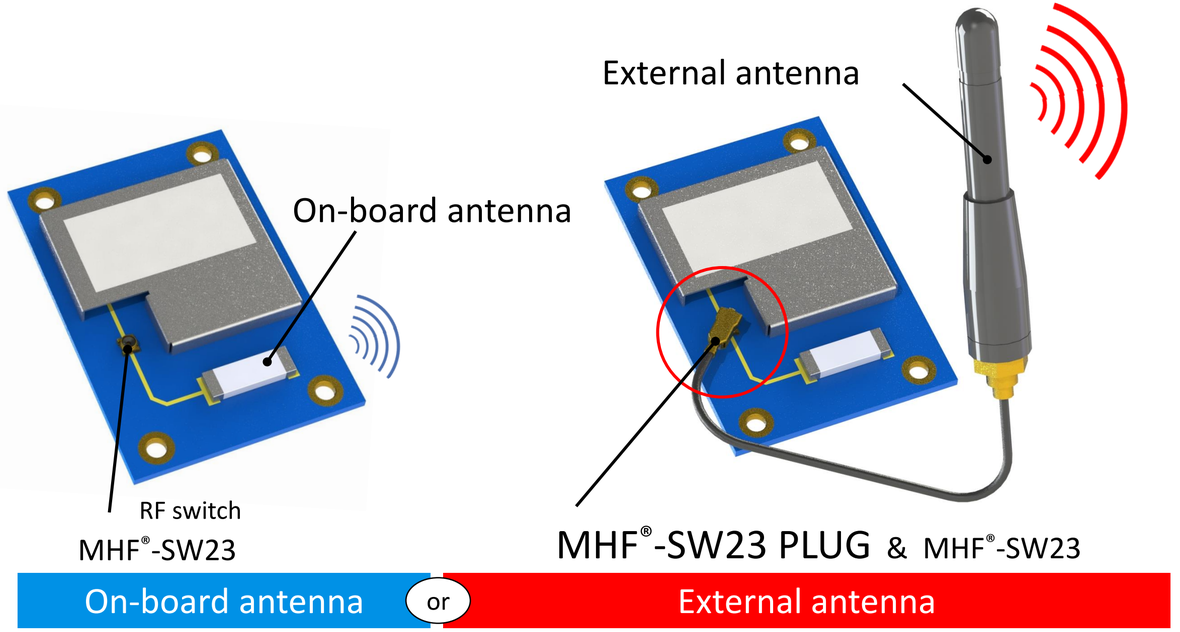
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Micro RF Coaxial Connector for RF switch : MHF®-SW23 plug

Click the photo below to get more detailed product information.

Product name		MHF®-SW23 Plug
Appearance		
		Mating with switch connector
Plug part number		20851-001R
Switch connector part number		20549-001E/-01 (MHF®-SW23)
Maximum height (mm)		3.7
Outside dimension of switch connector		2.3 x 2.3
Coax O.D. (Center Conductor AWG)		1.13 mm (32)
Frequency		DC - 6GHz
VSWR (L=100mm)	DC - 3GHz	1.4 max.
	3GHz - 6GHz	1.6 max.
Rated power		2W max.
Service temp. (Celsius)		-40 degree ~ +85 degree
Characteristic impedance		50ohm
Contact resistance	Signal contact	100m ohm max.
	Ground contact	100m ohm max.
Insulation resistance		DC100 1000M ohm min :Initial 10M ohm min :After test
Withstand voltage		AC200V/min

Flexibility to choose on-board antenna or external antenna



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2.4 mm RF CONNECTOR

Click the photos below to get more detailed product information.

Product name	2.4 mm RF CONNECTOR CONVERSION ADAPTER PLUG TO PLUG	2.4 mm RF CONNECTOR CONVERSION ADAPTER JACK TO JACK	2.4 mm RF CONNECTOR CONVERSION ADAPTER PLUG TO JACK	2.4 mm RF CONNECTOR END LAUNCH ADAPTER FOR PCB (PLUG)	2.4 mm RF CONNECTOR END LAUNCH ADAPTER FOR PCB (JACK)
Appearance					
Part number	50201-0001	50201-JJ01	50201-0J01	50202-0001	50202-J001
Type	Conversion adapter			End Launch adapter	
	Plug to Plug	Jack to Jack	Plug to Jack	Plug	Jack
Frequency	DC～50GHz				

Product name	2.4 mm RF CONNECTOR ADAPTER FOR PCB (INNER LAYER)	2.4 mm RF CONNECTOR ADAPTER FOR PCB (OUTER LAYER)	2.4 mm RF CONNECTOR TERMINATION ADAPTER (PLUG)	2.4 mm RF CONNECTOR TERMINATION ADAPTER (JACK)
Appearance				
Part number	50203-J001	50204-J001	50205-0001	50205-J001
Type	Panel mount adapter		Termination adapter(Nonreflective terminator)	
	Jack	Jack	Plug	Jack
Frequency	DC～50GHz			

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Appendix : I-PEX Connectors Series

Click icon below to see latest product information



MHF®

High-Frequency Micro RF Connectors



CABLINE® FPL®

High-Speed Micro-Coaxial Connectors



NOVASTACK®

High-Density Board-to-Board Connectors



EVAFLEX® MINIFLEX®

High-Density FFC/FPC Connectors



ISH® / IARPB® AP-series

High-Power Wire-t -Board Board-to-Board Connectors

CUSTOM

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